COD DELIVERY RISK: OPERATIONAL REVIEW

Oytra | *Associate Software Engineer take-home assignment*

Executive summary

The supplied file contains 2,018 rows. All 2,018 rows are labelled COD. Among 2,017 orders with a final Delivered or RTO outcome, 573 are RTO and 1,444 are Delivered, giving an observed RTO rate of **28.4%**. Average order value among resolved orders: ■**919**.

Key observations

RTO order value totals ■547,264 (gross order value, not lost profit). 1 row(s) have a non-final/other status and are excluded from the RTO-rate denominator. Highest courier cohorts: Xpressbees 31.2% (n=397), Ecom Express 30.2% (n=305), BlueDart 28.9% (n=187). Highest acquisition-source cohorts: meta_ads 31.3% (n=779), instagram_organic 29.4% (n=235), google_ads 28.0% (n=246). These are descriptive comparisons, not evidence of causation.

Recommended operating playbook

1. Confirm before dispatch: send order ID, item summary and COD amount with Confirm/Cancel choices. **2. One reminder:** remind non-responders once; request landmark/PIN confirmation only when needed. **3. Human review:** use the risk score to prioritize review, never auto-reject customers. **4. Controlled pilot:** run a 2–4 week randomized pilot and measure confirmation, delivery, RTO, complaints and contribution margin. **5. Better data:** capture response, cancellation reason, delivery attempts, RTO reason and timestamps.

Illustrative upside — assumptions only

At a hypothetical 10% relative reduction in the 573 historical RTO count, approximately 57.3 RTOs might be avoided. Using average order value as a proxy gives ■52,660 associated order value. This is not profit or guaranteed savings; costs, margins and recoverability are unknown.

Mock WhatsApp templates

Confirmation: “Hi! Your order {{order_id}} for ■{{amount}} is placed with Cash on Delivery. Reply CONFIRM to proceed or CANCEL if you no longer need it. Thank you!”
Reminder: “A quick reminder about COD order {{order_id}}. Please confirm by {{time}} so we can prepare your shipment. Reply CONFIRM or CANCEL.” No API was connected and no messages were sent.

Limitations and next steps

Historical sample only; no response, shipping cost, product margin, delivery-attempt or RTO-reason fields. Risk score is a transparent heuristic, not a trained ML model: prior RTO +55; no prior orders +20; top-quartile value +15; 3+ items +10. Validate with operations and pilot before production. Do not use the score for automatic customer blocking.